**HKUST — MATH 2111 Matrix Algebra and Applications** **Sample C Solutions**
**Fall 2026–2027 Midterm Test** Total: 100 marks

**Qn. 1 (40 marks)**

(1) **(B).** The formula

$$T(x_1, x_2) = (x_1 - x_2,\ 2x_1 + 3x_2)$$

is homogeneous and additive. The other choices contain a translation, a product, or an absolute value.

(2) **(C).** If $[T] = A$ and $[S] = B$, then

$$[S \circ T] = [S][T] = BA.$$

(3) **(A).** If $Ax = 0$, then

$$x = I_3 x = BAx = 0.$$

Thus $\mathrm{Nul}(A) = \{0\}$, so the columns of $A$ are linearly independent.

(4) **(D).** Since $v_1, v_2$ form a basis of $\mathrm{Nul}(A)$, every solution is a particular solution plus an arbitrary homogeneous solution:

$$x = x_p + sv_1 + tv_2, \qquad s, t \in \mathbb{R}.$$

The condition $b \neq 0$ gives $x_p \notin \mathrm{Nul}(A)$, so the unshifted nullspace in choice (B) is not the solution set.

(5) **(C).** Since $A$ is $3 \times 3$,

$$\det(2A^{-1}) = 2^3 \det(A^{-1}) = 8\frac{1}{-4} = -2.$$

(6) **(D).** The $j$th column of $AB$ is $A$ times the $j$th column of $B$:

$$A\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 2a_1 - a_2 + 3a_3.$$

(7) **(B).** The relation is exactly

$$A\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = a_1 - 2a_2 + a_3 = 0.$$

(8) **(A).** Expanding along row 2,

$$\det\begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ 3 & 0 & 4 \end{bmatrix} = -\det\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = -5.$$

(9) **(C).** Taking determinants in $A^2 = 0$ gives

$$(\det A)^2 = \det 0 = 0,$$

so $\det A = 0$.

(10) **(D).** If $v \neq 0$ satisfied $Av = 0$ and $x_0$ solved $Ax = b_0$, then $x_0 + v$ would be a second solution. Hence the homogeneous system has only the zero solution.

| qn. | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
|---|---|---|---|---|---|---|---|---|---|---|
| ans. | B | C | A | D | C | D | B | A | C | D |

**Qn. 2 (14 marks) (a)** A common vector has two representations:

$$a_1u_1 + a_2u_2 + a_3u_3 = b_1w_1 + b_2w_2.$$

Comparing the four coordinates gives

$$\begin{aligned} a_1 + a_3 &= b_1 + b_2, \quad & a_2 + a_3 &= b_1 - b_2, \\ a_1 + a_2 &= 2b_2, & b_1 &= b_2. \end{aligned}$$

Put $b_1 = b_2 = t$. Then $a_2 + a_3 = 0$, while

$$a_1 + a_3 = 2t, \qquad a_1 + a_2 = 2t.$$

Thus $a_2 = a_3 = 0$ and $a_1 = 2t$. Hence every common vector is

$$2tu_1 = t(w_1 + w_2),$$

so

$$\boxed{U \cap W = \mathrm{Span}\{u_1\}.}$$

**(b)** Let $P = [\, u_1 \; u_2 \; u_3 \; w_1 \,]$. Then

$$P = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \det P = \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = -2 \neq 0.$$

Therefore $\{u_1, u_2, u_3, w_1\}$ is a basis of $\mathbb{R}^4$. For every $z \in \mathbb{R}^4$,

$$z = c_1u_1 + c_2u_2 + c_3u_3 + c_4w_1 = \underbrace{(c_1u_1 + c_2u_2 + c_3u_3)}_{\in U} + \underbrace{c_4w_1}_{\in W}.$$

**(c)** If $z = u + w = u_0 + w_0$, then

$$u - u_0 = -(w - w_0) \in U \cap W = \mathrm{Span}\{u_1\}.$$

Hence, for some $t \in \mathbb{R}$,

$$\boxed{u = u_0 + tu_1, \qquad w = w_0 - tu_1.}$$

Conversely every such pair gives $z$. The decomposition is not unique because $t$ is arbitrary.

**Qn. 3 (16 marks) (a)** Since $u_3 = u_1 + u_2$, linearity requires

$$T(u_3) = T(u_1) + T(u_2) = e_1 + e_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Thus

$$\boxed{p = 1, \qquad q = 0.}$$

These values are sufficient because $u_1, u_2, e_3$ form a basis, and $T(e_3)$ may then be chosen arbitrarily.

**(b)** Write, with $a, b, c \in \mathbb{R}$,

$$z = T(e_3) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Since $e_1 = u_1 - e_3$ and $e_2 = u_2 - e_3$,

$$T(e_1) = e_1 - z, \qquad T(e_2) = e_2 - z.$$

Therefore every possible standard matrix is

$$\boxed{A(a,b,c) = \begin{bmatrix} 1-a & -a & a \\ -b & 1-b & b \\ -c & -c & c \end{bmatrix}.}$$

Adding column 3 to columns 1 and 2 gives $[\, e_1 \; e_2 \; z \,]$, so

$$\det A(a,b,c) = \det[\, e_1 \; e_2 \; z \,] = c.$$

Hence

$$\boxed{T \text{ is invertible exactly when } c \neq 0.}$$

**Qn. 3 (continued)**
**(c)** Suppose $T^2 = T$. Applying this identity to $u_1$ gives

$$T^2(u_1) = T(e_1) = e_1 - z = T(u_1) = e_1,$$

so $z = 0$, i.e. $a = b = c = 0$. Conversely, this choice gives

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad A^2 = A.$$

Thus it is the unique transformation satisfying $T^2 = T$. Its fixed vectors solve

$$Ax = x \iff x_3 = 0,$$

so

$$\boxed{\{x : T(x) = x\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\}.}$$

**Qn. 4 (15 marks) (a)** The standard matrices are

$$C = [R] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}, \qquad B = [S] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Therefore

$$A = [T] = BC = \boxed{\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 \end{bmatrix}}.$$

**(b)** The map $S$ is one-to-one, because $S(a, b) = 0$ forces $a = b = 0$. Hence

$$T(x) = 0 \iff R(x) = 0.$$

Thus

$$x_1 + x_3 = 0, \qquad x_2 + x_3 + x_4 = 0.$$

Let $x_3 = s$ and $x_4 = t$. Then

$$\boxed{x = s \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.}$$

**(c)** The map $R$ is onto $\mathbb{R}^2$, since $R(a, b, 0, 0) = (a, b)$. Hence the possible outputs of $T = S \circ R$ are exactly the outputs of $S$:

$$\boxed{\text{Range}(T) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} : y_3 = y_1 + y_2 \right\}.}$$

Part (b) gives nonzero vectors in the kernel, so $T$ is not one-to-one. Its range is a plane, not all of $\mathbb{R}^3$, so $T$ is not onto $\mathbb{R}^3$.

**Qn. 5 (15 marks) (a)** Since the columns are the images of $e_1, e_2, e_3$,

$$A_\alpha = \begin{bmatrix} 1+\alpha & \alpha & \alpha \\ \alpha & 1+\alpha & \alpha \\ \alpha & \alpha & 1+\alpha \end{bmatrix}.$$

Apply

$$R_2 \leftarrow R_2 - R_1, \qquad R_3 \leftarrow R_3 - R_1,$$

then

$$R_1 \leftarrow R_1 - \alpha R_2 - \alpha R_3.$$

This gives

$$\begin{bmatrix} 1+3\alpha & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix},$$

so

$$\boxed{\det A_\alpha = 1 + 3\alpha.}$$

**(b)** Therefore $T_\alpha$ is invertible exactly when

$$\boxed{\alpha \neq -\tfrac{1}{3}.}$$

Fix such an $\alpha$. Let $s = x_1 + x_2 + x_3$ and $\mathbf{1} = [\,1\ 1\ 1\,]^T$. Since the coordinate sum of $T_\beta(x) = x + \beta s\mathbf{1}$ is $(1+3\beta)s$,

$$\begin{aligned} T_\alpha(T_\beta(x)) &= x + \beta s\mathbf{1} + \alpha(1+3\beta)s\mathbf{1} \\ &= x + (\alpha + \beta + 3\alpha\beta)s\mathbf{1}. \end{aligned}$$

Thus

$$\boxed{T_\alpha \circ T_\beta = T_{\alpha+\beta+3\alpha\beta}.}$$

For the identity transformation, the parameter must be 0, so

$$\alpha + \beta + 3\alpha\beta = 0 \quad \Longrightarrow \quad \boxed{\beta = -\frac{\alpha}{1+3\alpha}.}$$

Because the composition parameter $\alpha + \beta + 3\alpha\beta$ is symmetric in $\alpha$ and $\beta$, the same choice also gives $T_\beta \circ T_\alpha = T_0$. Hence

$$\boxed{T_\alpha^{-1} = T_{-\alpha/(1+3\alpha)}}.$$

**(c)** Let $T_{-1/3}(x) = b$ and put $s = x_1 + x_2 + x_3$. Then

$$b = x - \frac{s}{3}\mathbf{1}.$$

Taking coordinate sums gives the necessary condition

$$\boxed{b_1 + b_2 + b_3 = 0.}$$

Conversely, if this condition holds, then for every $t \in \mathbb{R}$,

$$x = b + t\mathbf{1}$$

has coordinate sum $3t$ and satisfies

$$T_{-1/3}(x) = b + t\mathbf{1} - t\mathbf{1} = b.$$

These are all solutions, because any solution satisfies $x - b = (s/3)\mathbf{1}$; set $t = s/3$. Therefore

$$\boxed{x = b + t\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R},}$$

provided $b_1 + b_2 + b_3 = 0$; otherwise there is no solution.

*End of solutions*